

Rotation operator part 1

Let $R_z(\alpha)$ be the rotation operator

$$R_z(\alpha) = \exp\left(\frac{i\alpha}{\hbar}L_z\right) = \exp\left(\alpha\frac{\partial}{\partial\phi}\right)$$

Show that

$$R_z(\alpha)Y_{lm}(\theta, \phi) = Y_{lm}(\theta, \phi + \alpha) \quad (1)$$

For spherical harmonic $Y_{lm}(\theta, \phi)$ we have

$$Y_{lm}(\theta, \phi) = (-1)^m \sqrt{\frac{(2l+1)(l-m)!}{4\pi(l+m)!}} P_l^m(\cos\theta) e^{im\phi}$$

Noting that $e^{im\phi}$ is the only factor that depends on ϕ

$$\begin{aligned} R_z(\alpha)e^{im\phi} &= \left(1 + \alpha\frac{\partial}{\partial\phi} + \frac{\alpha^2}{2!}\frac{\partial^2}{\partial\phi^2} + \frac{\alpha^3}{3!}\frac{\partial^3}{\partial\phi^3} + \dots\right) e^{im\phi} \\ &= \left(1 + im\alpha + \frac{(im\alpha)^2}{2!} + \frac{(im\alpha)^3}{3!} + \dots\right) e^{im\phi} \\ &= e^{im\alpha} e^{im\phi} \end{aligned}$$

Hence

$$R_z(\alpha)Y_{lm}(\theta, \phi) = Y_{lm}(\theta, \phi)e^{im\alpha} = Y_{lm}(\theta, \phi + \alpha)$$

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